RAILWISE: RAILWAY PLATFORM ALLOCATION AI AGENT

ASSIGNMENT: AI AGENT FORMULATION AND ANALYSIS

1. Agent Overview and Problem Identification

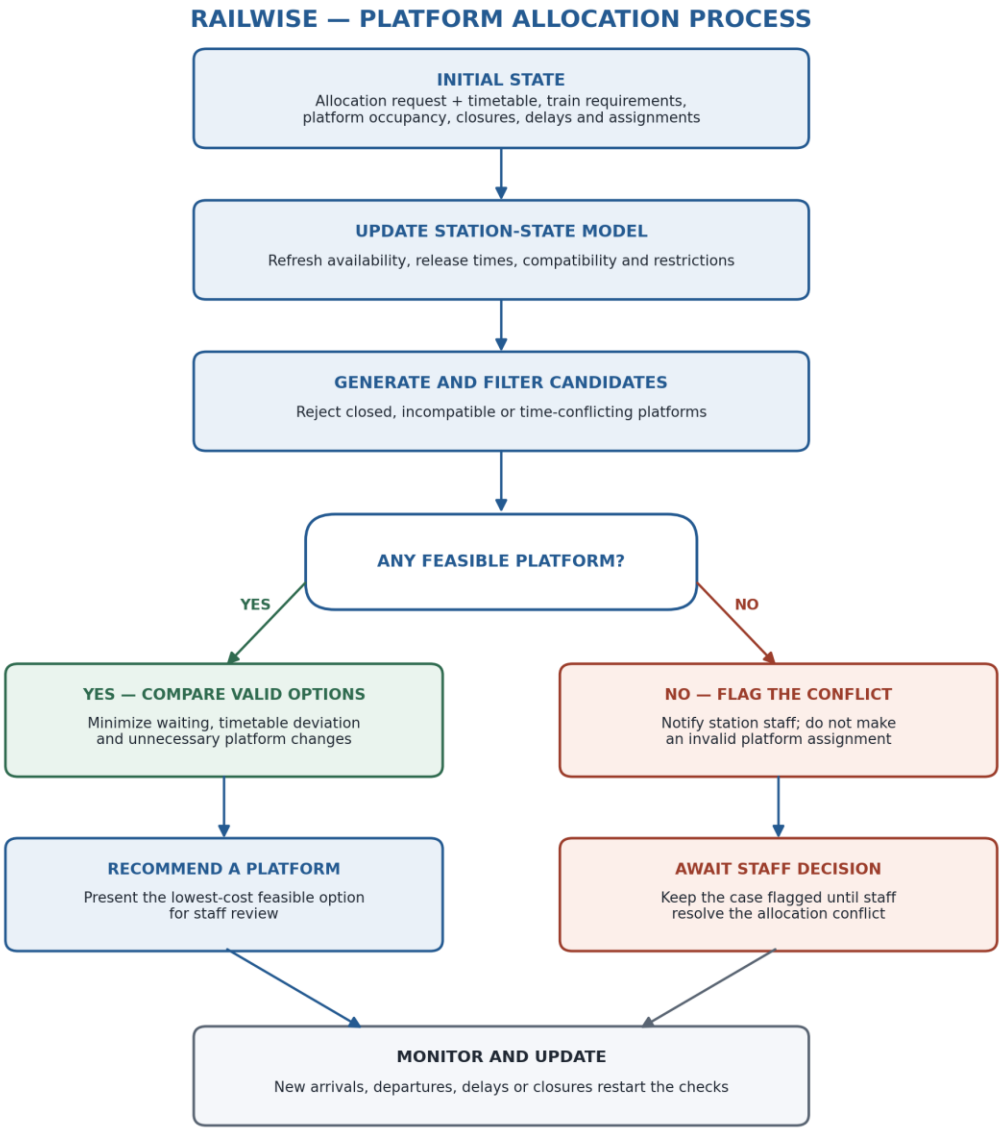
RailWise is a decision-support AI agent that recommends suitable platforms for arriving and departing trains. It considers train schedules, platform occupancy and release times, train requirements, delays, closures, and operating restrictions. Its goal is to reduce platform conflicts, avoid unnecessary waiting and reallocations, and flag situations that require staff review.

2. Agent Type

RailWise is a model-based, utility-based agent. It maintains an updated representation of station conditions, filters out assignments that violate hard constraints, and compares the remaining feasible options. Among valid options, it prefers lower waiting time, less timetable deviation, and fewer unnecessary changes.

3. Graphical Representation

The top-down flowchart shows how RailWise checks platform constraints, handles feasible and infeasible cases, and revisits decisions when station conditions change.



4. Problem Formulation

The agent checks hard constraints before comparing the cost of valid options. If no feasible platform exists, it does not force an assignment.

INITIAL STATE | Train allocation request received

Read timetable, train requirements, occupancy, release times, closures and delays

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STATE 2 | Update station-state model

Generate candidates and reject incompatible, closed, or time-conflicting platforms

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DECISION | Is at least one feasible platform available?

YES — FEASIBLE	NO — INFEASIBLE
Compare valid options by waiting time, timetable deviation, and unnecessary changes.\n↓\nRECOMMENDATION: Propose the lowest-cost valid platform.	Do not assign an invalid platform.\n↓\nSTAFF REVIEW: Flag conflict and await a human decision.

FEEDBACK LOOP: New arrivals, departures, delays, or closures update the model and restart checks.

Operational boundary: RailWise supports station staff; it does not operate signals or authorize train movements.

5. PEAS Framework

PEAS component	RailWise description
Performance measure	Avoid platform conflicts; respect restrictions; reduce waiting, timetable deviation, and unnecessary reallocations; flag infeasible cases.
Environment	Railway station, trains, platforms, timetables, station staff, and changing operating conditions.
Actuators	Proposed platform assignments, staff alerts, schedule displays, and allocation reports.
Sensors	Timetables, train requirements, arrival/departure updates, platform occupancy, closures, and delay reports.

6. Environmental Properties

Property	Classification	Reason
Observability	**Partially observable**	Updates can be delayed or incomplete.
Determinism	**Stochastic**	Unexpected delays and operational changes affect outcomes.
Episodes	**Sequential**	One assignment affects later allocation choices.
Change	**Dynamic**	Schedules, occupancy, and restrictions can change during operation.
Values	**Hybrid**	Platform choices are discrete; times and delays are numerical.
Agents	**Multi-agent**	Staff, drivers, and railway control systems also affect operations.

7. Example Scenario

Train T12 is due at 10:20. Platform 1 remains occupied until 10:25, Platform 3 is closed, and Platform 2 is free. RailWise rejects Platforms 1 and 3, checks Platform 2 for compatibility and timing, and recommends it if all constraints are satisfied. If no platform is feasible, it flags the conflict for staff review.

Conclusion

RailWise is a model-based, utility-based decision-support agent for sequential railway platform allocation. It prioritizes feasible assignments, accounts for changing and uncertain conditions, and leaves unresolved conflicts to station staff.